

Assignment Summary:

The objective of the Clustering of Countries assignment was to help the CEO of the GIVE Foundation identify the countries that are in the greatest need of financial aid. The NGO had raised approximately \$10 million and needed a data-driven approach to decide where the funds should be allocated. The problem was solved by grouping countries based on economic and health indicators so that underdeveloped countries could be clearly identified.

The analysis started with data inspection and exploratory data analysis (EDA). This included checking the structure of the dataset, identifying missing values, and understanding the distribution of variables. Outlier analysis was performed to detect extreme values that could affect clustering results.

Before applying clustering algorithms, the data was scaled using standardization so that all variables contributed equally to the model. Two clustering techniques were tested: K-Means Clustering and Hierarchical Clustering (using different linkages). The optimal number of clusters for K-Means was determined using the Elbow Method, which helped identify a suitable value of k .

After comparing the results of both methods, K-Means clustering produced clearer and more interpretable clusters, making it more suitable for this problem. The clusters were analyzed by comparing key development indicators such as GDP per capita, income, and child mortality rate. The cluster with low GDP, low income, and high child mortality represented the least developed countries. These countries were identified as the ones that should be prioritized for aid and were recommended to the CEO as the final list of countries requiring immediate support.

Clustering:

a) Compare and contrast K-Means Clustering and Hierarchical Clustering

K-Means Clustering and Hierarchical Clustering are two widely used unsupervised machine learning techniques used to group similar data points into clusters.

K-Means Clustering is a partition-based algorithm that divides the dataset into a predefined number of clusters (k). It works by assigning data points to the nearest cluster centroid and iteratively updating the centroids until convergence. K-Means is fast in execution and works well for large datasets. However, it requires the number of clusters to be specified in advance and is sensitive to outliers and the initial centroid positions.

In contrast, Hierarchical Clustering builds a tree-like structure called a dendrogram, which represents the nested grouping of data points. It does not require specifying the number of clusters initially. Clusters can be formed either by Agglomerative (bottom-up) approach, where each data point starts as its own cluster and merges step by step, or Divisive (top-down) approach, where one large cluster is recursively split.

The key difference is that K-Means produces flat clusters, while Hierarchical Clustering produces hierarchical relationships between clusters. K-Means is faster and suitable for large datasets, whereas Hierarchical Clustering requires more processing power but is useful for understanding the structure of the data.

b) Steps of the K-Means Clustering Algorithm

- Select the number of clusters (k) based on the problem or statistical methods.
- Initialize centroids randomly by selecting k data points from the dataset.
- Assign data points to clusters by calculating the distance (usually Euclidean distance) between each data point and the cluster centroids. Each point is assigned to the nearest centroid.
- Update cluster centroids by computing the mean of all data points assigned to each cluster.
- Repeat the assignment and update steps until the centroids no longer change significantly or the algorithm reaches convergence.
- The algorithm aims to minimize the within-cluster sum of squares (WCSS), which represents the total distance between data points and their respective cluster centroids.

c) How is the value of 'k' chosen in K-Means clustering?

The value of k can be chosen using both statistical techniques and business understanding.

From a statistical perspective, the most common method is the Elbow Method. In this method, the Within-Cluster Sum of Squares (WCSS) is calculated for different values of k . A graph is plotted between the number of clusters and the WCSS. Initially, WCSS decreases rapidly as k increases, but after a certain point the decrease slows down. The point where the curve forms an "elbow" is considered the optimal number of clusters.

From a business perspective, the selected value of k should make practical sense for the problem being solved. In this assignment, the objective was to classify countries based on development levels. Therefore, clusters should represent meaningful groups such as developed, developing, and underdeveloped countries so that the organization can easily decide where to allocate funds.

d) Necessity of Scaling / Standardization before Clustering

Scaling or standardization is an important preprocessing step before applying clustering algorithms.

Clustering algorithms such as K-Means use distance-based calculations (commonly Euclidean distance) to group similar data points. If variables have different scales or units, variables with larger numeric ranges will dominate the distance calculation.

For example, in the country dataset, GDP per capita may have values in thousands, while child mortality rate may have values in tens. Without scaling, the GDP variable would have a much larger influence on the clustering result, causing the algorithm to ignore other important variables.

Standardization transforms the variables so that they have mean = 0 and standard deviation = 1. This ensures that all features contribute equally to the clustering process.

By performing scaling, we improve the accuracy, stability, and reliability of the clustering results.

e) Different Linkages used in Hierarchical Clustering

In Hierarchical Clustering, linkage methods determine how the distance between clusters is calculated when merging clusters.

The main linkage methods include:

1. Single Linkage (Nearest Neighbor): The distance between two clusters is defined as the minimum distance between any two points from the clusters. This method can sometimes produce elongated clusters due to the chaining effect.
2. Complete Linkage (Farthest Neighbor): The distance between clusters is calculated as the maximum distance between any two points in the clusters. This method produces more compact clusters compared to single linkage.